

End-to-end neural compensation for target deformation and lens distortion in camera calibration

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ABSTRACT

High-precision camera calibration is often constrained by distortion modeling errors and target deviations. Based on our previous depth-dependent compensation model, we have advanced this distortion model by eliminating the requirement for depth acquisition. Previously, the operational complexity of obtaining depth information hindered the practical usability of the algorithm. At the same time, standard calibration targets often introduce geometric uncertainties due to manufacturing tolerances. To resolve these limitations in practical deployment and hardware accuracy, we propose TLNet, a physics-guided neural network framework. By integrating differentiable physical projection with neural compensation modules, TLNet simultaneously optimizes for three-dimensional target deformation and two-dimensional residual distortion. This end-to-end joint optimization ensures global consistency between the spatial and optical models, achieving highly accurate calibration that are robust to subtle target imperfections. Experimental results demonstrate that this method provides absolute accuracy and algorithmic stability across different focal lengths and dataset scales, establishing a reliable geometric foundation for advanced 3D visual perception.

1. Introduction

Camera calibration is the fundamental process of establishing a rigorous mathematical mapping between the three-dimensional physical world and the two-dimensional image plane. By accurately determining the intrinsic optical characteristics and extrinsic spatial poses of a camera, this process bridges the gap between raw pixel data and metric spatial information. Consequently, it serves as the critical prerequisite for a wide range of downstream applications that demand rigorous spatial accuracy, including high-resolution point cloud measurement [1] and three-dimensional scene reconstruction [2–4]. Recent advances in multi-camera and generalized relative pose estimation, such as six-point solvers and affine-correspondence-based methods, further emphasize the role of accurate camera calibration in broader multi-view vision systems [5–7].

Over the past decades, numerous calibration paradigms have been developed to achieve this high-fidelity geometric transformation. Early approaches including the Direct Linear Transformation [8] provided foundational mathematical frameworks but lacked distortion modeling.

Subsequently, Tsai's landmark method [9] introduced rigorous physical constraints for radial distortion. However, such highly accurate physical models heavily relied on elaborate three-dimensional calibration rigs. Alternatively, pioneer self-calibration techniques proposed by Maybank and Faugeras [10] attempted to deduce parameters from natural scene correspondences using Kruppa equations. While eliminating the need for physical targets, these approaches often suffered from algorithmic instability and low robustness. Striking an optimal balance between accuracy and operational convenience, Zhang's method [11] emerged as a revolutionary paradigm. By requiring only a few images of a planar pattern captured from freely moving viewpoints, it drastically simplified the physical setup and established itself as the ubiquitous industry benchmark. However, the classic mathematical model underlying such planar methods is built upon a strict physical premise. It demands the calibration target to be an absolutely rigid and perfect mathematical plane.

In practical industrial settings, calibration targets inevitably deviate from the ideal planar assumption due to manufacturing tolerances and material flexibility. To address this physical reality, several approaches

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estimate the three-dimensional board deformation alongside camera parameters [12–16]. Huang's method [16] provides a representative solution by introducing a set of structure parameters to describe the non-planar variations of the target. By incorporating these structural variables into a global bundle adjustment, the framework refines and recovers the actual three-dimensional geometry of the checkerboard to mitigate systematic errors caused by board warping. However, while Huang's method successfully reconstructs the target geometry, its capacity to correct two-dimensional projection errors remains restricted by classic mathematical polynomial models. These rigid parametric formulations struggle to capture complex, high-frequency manufacturing defects or severe nonlinear aberrations frequently encountered in low-cost and wide-angle lenses.

To overcome the representational limits of mathematical distortion models and traditional geometric paradigms, recent advancements introduce deep learning techniques to the optical calibration process. DeepCalib estimates intrinsic parameters for wide field-of-view cameras from a single image [17], and Lopez et al. further investigate single-image camera calibration with radial distortion [18]. Wakai et al. estimate camera orientation and remove fisheye distortion under the Manhattan world assumption using heatmap regression [19], while GeoCalib combines learning-based single-image calibration with geometric optimization [20]. These studies demonstrate the potential of neural networks in camera geometry estimation from natural or single-image observations. Deep learning has also been introduced into specific components of the calibration pipeline. MATE employs machine learning for adaptive calibration template detection [21], FisheyeRecNet learns image-level correction mappings for fisheye image rectification [22], and recent checkerboard detection methods improve feature localization using YOLOX or physics-informed neural networks [23,24]. These methods improve the robustness of image rectification, calibration pattern detection or feature localization, providing useful support for practical calibration tasks. Furthermore, among various neural architectures, learning-based methods with physical or geometric constraints are attracting increasing attention. Self-calibrating NeRF integrates camera parameter estimation into neural scene representation [25], while Li et al. reformulate the classic camera model as physical constraint equations incorporated into a neural-network-based calibration framework [26]. Compared with purely data-driven networks whose intermediate representations are often implicit, physics-guided methods provide a more interpretable way to combine neural compensation with physical camera modeling.

Although the above studies have advanced learning-based camera calibration from different perspectives, the joint modeling of physical target deformation and residual lens distortion remains insufficiently explored. In practical high-precision calibration, these two error sources are coupled through the projection process: target deformation introduces three-dimensional geometric deviations, while residual lens distortion produces image-plane projection errors. Existing neural approaches usually focus on intrinsic estimation, image rectification, feature localization or scene-level self-calibration. A complete physics-guided neural framework that simultaneously compensates for three-dimensional target deformation and two-dimensional residual lens distortion is still lacking.

In our previous work [27] we proposed a hybrid architecture to address complex residual distortions. We utilized spatial depth as an auxiliary input to guide a neural network in learning highly nonlinear projection errors. While numerically effective for precise distortion mapping this explicit depth coupling inevitably introduces additional operational overhead. Demanding physical depth data diminishes the operational simplicity of classic planar methods. This practical constraint motivates us to develop a more efficient calibration paradigm that eliminates the reliance on spatial depth inputs.

In this paper we propose TLNet which is a unified neural and physical framework for high precision camera calibration. The main contributions of our work are summarized as follows. First, we design an end

to end joint optimization mechanism to simultaneously compensate for three dimensional target deformation and two dimensional residual distortion. Second, by integrating explicit physical guidance the proposed framework ensures that the optimized network parameters retain strict physical meanings specifically corresponding to the actual camera intrinsics and extrinsics. Third, comprehensive experiments demonstrate that TLNet achieves superior accuracy and algorithmic stability across different focal lengths and limited dataset scales.

2. Related work

2.1. Classical camera calibration

Classical camera calibration forms the theoretical cornerstone of modern 3D computer vision, aiming to establish an accurate mapping from 3D spatial coordinates to 2D image pixels. The mainstream analytical paradigm, fundamentally represented by Zhang's method [11], relies on the pinhole camera model and planar calibration targets [28,29].

Mathematically, given a 3D point in the world coordinate system $\mathbf{P}_w = [X_w, Y_w, Z_w]^T$, the rigid transformation to the camera coordinate system $\mathbf{P}_c = [X_c, Y_c, Z_c]^T$ is parameterized by the extrinsic rotation matrix $\mathbf{R}$ and translation vector:

$$\mathbf{P}_c = \mathbf{R}\mathbf{P}_w + \mathbf{t} \quad (1)$$

The point is then projected onto the normalized image plane as $\mathbf{p}_n = [X_c/Z_c, Y_c/Z_c]^T$. To account for lens imperfections, the classical paradigm integrates parametric distortion models, such as the Brown-Conrady model [30], to map $\mathbf{p}_n$ to the distorted coordinates $\mathbf{p}_d$. Finally, the affine transformation to the pixel coordinate system $\mathbf{p}_{pixel} = [u, v]^T$ is formulated using the intrinsic matrix $\mathbf{K}$:

$$\mathbf{p}_{pixel} = \mathbf{K}\mathbf{p}_d \quad (2)$$

To solve for these parameters, the calibration process typically formulates a non-linear least-squares optimization problem. The objective is to minimize the global reprojection error between the predicted pixel coordinates and the actual observed feature points $\mathbf{p}_{obs}$:

$$\arg \min_{\mathbf{K}, \mathbf{R}, \mathbf{t}, \mathbf{D}} \sum_{i=1}^N \sum_{j=1}^M \left\| \mathbf{p}_{obs}^{i,j} - \Pi(\mathbf{K}, \mathbf{R}_i, \mathbf{t}_i, \mathbf{D}, \mathbf{P}_w^j) \right\|^2 \quad (3)$$

where $\Pi(\cdot)$ denotes the comprehensive projection function, $\mathbf{D}$ represents the distortion coefficients, and N and M are the number of images and target points. This objective is optimized utilizing the Levenberg-Marquardt (LM) algorithm [31].

However, this classical paradigm is fundamentally bounded by two implicit assumptions. First, it assumes the absolute geometric ideality of the target points $\mathbf{P}_w$, ignoring 3D spatial perturbations induced by manufacturing tolerances or thermal deformations [14]. Second, the representational capacity of polynomial distortion models $\mathbf{D}$ is insufficient for fitting complex, high-order, or non-radial residual distortions [32]. Consequently, uncompensated 3D geometric noise is propagated into the 2D pixel domain during the LM optimization, intrinsically limiting the upper bound of overall calibration accuracy.

2.2. Imperfect target calibration

To overcome the rigid planar assumption of classical methods, researchers have explored calibration techniques that account for physical target deformations [33]. A representative approach is the constrained optimization paradigm proposed by Huang et al. [16]. Recognizing that real-world calibration boards are rarely flat, this method simultaneously estimates the camera parameters and the actual 3D geometry of the target. The core of this discrete iterative process minimizes a global reprojection error over N images and M feature points:

$$\arg \min_{\mathbf{R}, \mathbf{t}, \mathbf{K}, \mathbf{D}, \mathbf{P}_w} \sum_{i=1}^N \sum_{j=1}^M \left\| \mathbf{p}_{obs}^{i,j} - \Pi(\mathbf{K}, \mathbf{R}_i, \mathbf{t}_i, \mathbf{D}, \mathbf{P}_w^j) \right\|^2 \quad (4)$$

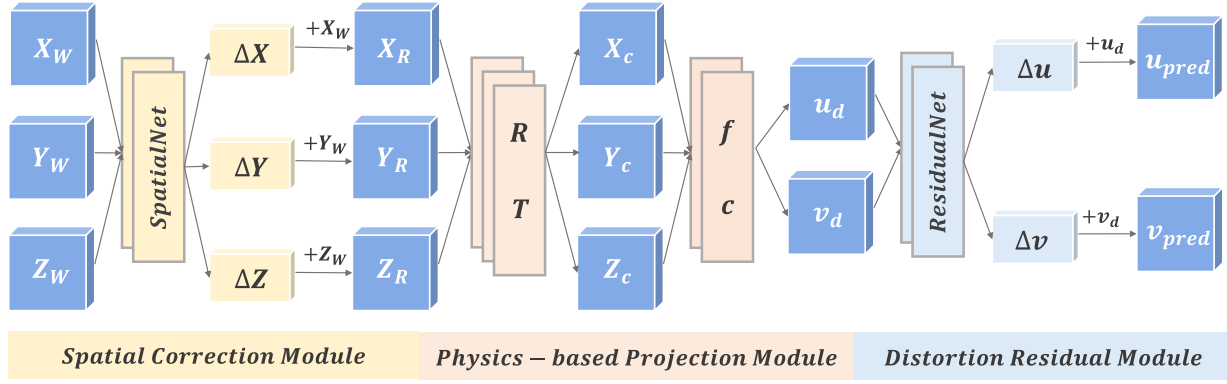


Fig. 1. Architecture of the proposed TLNet framework.

To resolve the severe parameter ambiguity between camera extrinsics and the optimized high-dimensional spatial coordinates $\hat{\mathbf{P}}_w^j$, this method applies statistical group constraints based on the nominal pattern design $\mathbf{P}_w^j$.

Specifically, the translation ambiguity is eliminated by forcing the optimized geometric center to coincide with the designed model:

$$\begin{cases} \sum_{j=1}^M \hat{X}_w^j = \sum_{j=1}^M X_w^j, \\ \sum_{j=1}^M \hat{Y}_w^j = \sum_{j=1}^M Y_w^j, \\ \sum_{j=1}^M \hat{Z}_w^j = \sum_{j=1}^M Z_w^j. \end{cases} \quad (5)$$

Rotational ambiguity is resolved by aligning the fitted surface normal to the Z_w -axis and restricting rotation around it:

$$\sum_{j=1}^M \left(\hat{X}_w^j Y_w^j - X_w^j \hat{Y}_w^j \right) \rightarrow 0 \quad (6)$$

Finally, a scale constraint is enforced to maintain the absolute dimensions in a statistical sense:

$$\sum_{j=1}^M \left\| \hat{\mathbf{P}}_w^j - \mathbf{P}_w^j \right\|^2 \rightarrow 0 \quad (7)$$

By enforcing these statistical attributes, the strategy recovers the undulating 3D shape of an imperfect target without requiring prior physical measurements [34].

However, despite its advancement in 3D spatial refinement, this approach faces inherent limitations when handling complex optical aberrations. It relies on a discrete optimization architecture that couples the 3D spatial deformation and 2D lens distortion within a single cost function. More importantly, its capability to compensate for lens distortion remains strictly bounded by conventional mathematical models. Because standard polynomial formulas possess limited representational capacity, this method struggles to accurately capture complex residual distortions [35].

3. Method

3.1. Overview

In our previous work [27], we analyzed the physical correlation between spatial depth and optical aberration. We discovered that optical aberration exhibits a strong dependence on spatial depth. To address this correlation, we proposed a depth-aware neural compensation strategy.

While numerically effective, this depth-coupling introduces significant operational overhead during the calibration process, limiting the algorithm's practical usability. Therefore, removing the Z-dependency from the 2D distortion model becomes a practical necessity. Concurrently, from a hardware perspective, standard calibration boards suffer from manufacturing tolerances. Practitioners often utilize low-fidelity alternatives such as printed paper or computer monitors to prioritize convenience and cost. These practical targets lack spatial precision, exhibiting inevitable out-of-plane geometric fluctuations along the Z-axis. To resolve this algorithmic limitation and the physical reality of imperfect hardware, we propose a paradigm shift towards an end-to-end calibration approach.

The central innovation of our methodology is a unified neural-physical framework, termed the Target and Lens Network (TLNet). As the name implies, it is designed to simultaneously compensate for 3D target deformation and 2D lens distortion. Traditional approaches like Zhang's method rely on a rigid planar assumption, completely ignoring 3D target deformation. While Huang's method addresses this by estimating the actual target geometry, its capability to handle 2D distortion is still restricted by conventional mathematical models. TLNet overcomes these limitations by integrating both 3D spatial correction and 2D distortion mapping into a single differentiable pipeline. As shown in Fig. 1, this end-to-end design allows the error signal from the final prediction to flow back through the entire network, ensuring that all parameters are optimized in a global perspective.

The framework consists of three interconnected modules designed to operate within a unified computational graph. The Spatial Correction Module first processes raw world coordinates through a lightweight MLP to map them into a refined 3D geometric space. This continuous spatial field is then fed into the Physics-based Projection Module, which executes differentiable pinhole transformations to project the coordinates onto the image plane. Finally, the Distortion Residual Module compensates for high-frequency optical misalignments in the 2D domain. Because these modules are differentiable and linked end-to-end, the error signal from the final pixel prediction propagates back through the physical backbone to the 3D input, allowing the spatial and residual parameters to be optimized collaboratively.

3.2. Spatial correction module

The TLNet pipeline begins with the Neural Spatial Correction Module, as detailed in Fig. 2. This component is designed to resolve 3D geometric inconsistencies of the calibration target. Standard calibration methods usually rely on a rigid planar assumption, which treats the checkerboard as an ideal Euclidean plane where all feature points are constrained to a nominal depth of $Z_w = 0$. In real-world scenarios, calibration targets are often affected by physical imperfections such as manufacturing tolerances, material thermal expansion, or structural

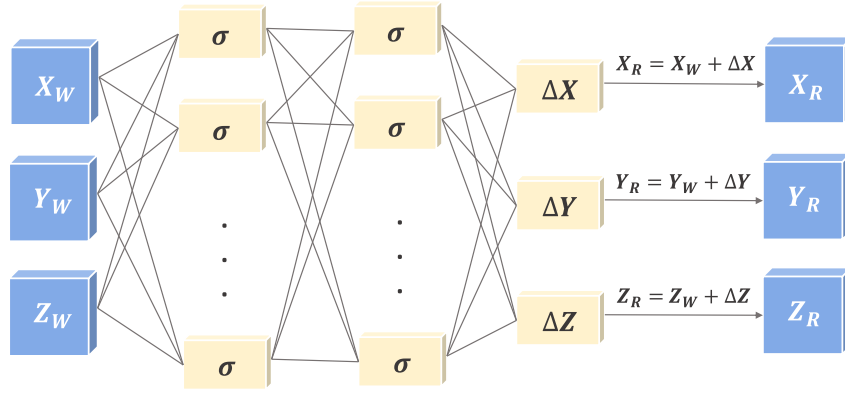


Fig. 2. Architecture of the spatial correction module.

warping. When these 3D deviations are present, a standard projection model is forced to absorb these geometric errors into the 2D lens distortion parameters during optimization. This results in a fundamental model mismatch where the estimated focal length and distortion coefficients lose their physical integrity.

To address these limitations, we model the geometric deviation of the target as a continuous spatial warping field. We implement this field using a lightweight Multi-Layer Perceptron (MLP), denoted as $F_{\text{spatial}}(\cdot; \theta_s)$. This neural approach treats the correction as a function of the position of the point. The module takes the raw world coordinates $\mathbf{P}_w = [X_w, Y_w, Z_w]^T$ as input. The network then learns a non-linear mapping to generate a 3D displacement vector $\Delta \mathbf{P} = [\Delta X, \Delta Y, \Delta Z]^T$. The refined world coordinates $\mathbf{P}_R$ are calculated by adding this displacement to the original coordinates:

$$\mathbf{P}_R = \mathbf{P}_w + F_{\text{spatial}}(\mathbf{P}_w; \theta_s) \quad (8)$$

where θ_s represents the learnable parameters of the spatial network. To ensure that the geometric correction remains physically plausible, we keep the MLP architecture lightweight and initialize the output weights to near-zero values. This ensures that $\mathbf{P}_R$ starts at the nominal positions at the beginning of training, allowing the framework to correct 3D deviations while maintaining the overall structure of the calibration target.

3.3. Physics-based projection module

Unlike the static mathematical model in classic calibration, the projection process in TLNet is implemented as a sequence of differentiable layers. This architecture ensures that the geometric constraints of the pinhole model are strictly maintained during the end-to-end optimization. Instead of re-estimating parameters in isolation, this module functions as a fixed physical backbone that propagates error signals from the image plane back to the spatial coordinates.

The refined world coordinates $\mathbf{P}_R$ are first transformed into the camera coordinate system $\mathbf{P}_c$ through a differentiable extrinsic layer. This layer performs a rigid body transformation using the current estimates of rotation $\mathbf{R}$ and translation $\mathbf{t}$:

$$\mathbf{P}_c = \mathbf{R}\mathbf{P}_R + \mathbf{t} \quad (9)$$

During training, $\mathbf{R}$ and $\mathbf{t}$ are treated as learnable parameters within the network. By representing the rotation through unit quaternions, we avoid the singularities common in Euler angle representations and ensure smooth gradient flow during backpropagation.

The subsequent projection stage maps the 3D points $\mathbf{P}_c = [x_c, y_c, z_c]^T$ onto the pixel coordinate system. This is achieved through a standardized projection operator that divides the spatial components by

the depth value z_c , followed by the application of the intrinsic matrix $\mathbf{K}$:

$$\begin{cases} u_d = f_x \frac{x_c}{z_c} + c_x, \\ v_d = f_y \frac{y_c}{z_c} + c_y. \end{cases} \quad (10)$$

By embedding these classic equations into a differentiable framework, the Physics-based Projection Module acts as a structural constraint. It forces the TLNet framework to find a solution that satisfies both the deep-learning-based corrections and the fundamental laws of optics. This integration ensures that the estimated intrinsic and extrinsic parameters retain their physical definitions while achieving higher numerical precision.

3.4. Distortion residual module

Following the physical projection, the TLNet pipeline employs the Distortion Residual Module to provide 2D pixel-level compensation, as shown in Fig. 3. While the Physics-based Projection Module handles the primary geometric transformation, it cannot fully account for high-frequency optical distortions or complex lens misalignments that stay outside the ideal pinhole model. To bridge this gap, we employ a second MLP, denoted as $F_{\text{resid}}(\cdot; \theta_r)$, to learn the pixel-level residual field.

To ensure stable gradient flow and prevent numerical saturation in the neural network, we apply a normalization step to the ideal pixel coordinates $\mathbf{p}_d = [u_d, v_d]^T$ obtained from the projection layer. These coordinates are scaled by a factor of 1000 to produce the standardized inputs $\mathbf{p}_n = [u_n, v_n]^T$. This numerical conditioning is essential because it keeps the input values within the sensitive region of the activation functions, thereby avoiding the gradient vanishing problem during the end-to-end training process.

Unlike our previous work [27] where the 3D depth Z was incorporated as an auxiliary input to the compensation network, TLNet relies on image-plane positions for residual mapping. While the inclusion of depth information can yield superior calibration numerals in controlled environments, it creates a practical bottleneck for subsequent 3D reconstruction tasks. Since the primary objective of reconstruction is to recover unknown spatial geometry from 2D observations, a residual model that is coupled with target depth becomes difficult to implement in generalized measurement scenarios. By removing this depth dependency, we ensure that the learned residual field is a pure representation of the camera internal optical characteristics. This strategic simplification enhances the versatility of TLNet, allowing the calibrated model to be integrated into various 3D reconstruction pipelines without requiring any prior knowledge of the scene geometry.

The normalized coordinates $\mathbf{p}_n$ are fed into the residual MLP which functions as a universal function approximator. Unlike traditional radial

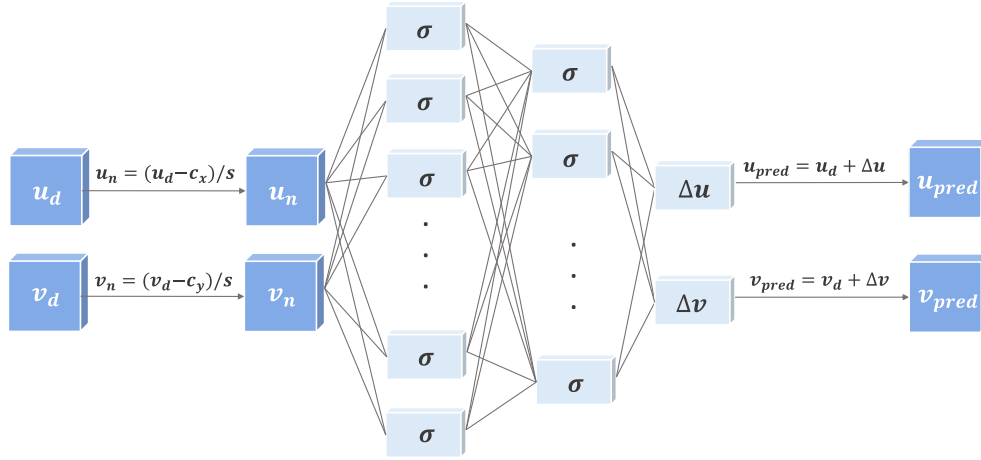


Fig. 3. Architecture of the distortion residual module.

and tangential distortion formulas that use a limited number of fixed coefficients, this neural mapping can capture arbitrary non-linear patterns across the image plane. The module predicts a 2D pixel displacement $\Delta \mathbf{p}$, and the final predicted pixel coordinates $\mathbf{p}_{pred}$ are formulated as:

$$\mathbf{p}_{pred} = \mathbf{p}_d + \mathcal{F}_{resid}(\mathbf{p}_n; \theta_r) \quad (11)$$

where $\mathbf{p}_d$ represents the ideal projection point from the physical backbone and θ_r denotes the learnable parameters of the residual network. This additive structure ensures that the physical model provides a stable global baseline while the neural network focuses exclusively on fine-tuning the local systematic errors. By integrating this module into the differentiable framework, TLNet achieves sub-pixel calibration accuracy that exceeds the performance of classic parametric models.

3.5. Loss function

The final stage of the TLNet framework involves the joint optimization of all learnable parameters, including the spatial correction weights θ_s , the extrinsic parameters $\{\mathbf{R}, \mathbf{t}\}$, the intrinsic matrix $\mathbf{K}$, and the residual network weights θ_r . To achieve sub-pixel accuracy while maintaining physical consistency, we define a total loss function L_{total} that guides the end-to-end training process.

The primary objective is to minimize the reprojection error L_{rep} , which is calculated as the mean squared error between the predicted pixel coordinates $\mathbf{p}_{pred,i}$ and the ground-truth feature points $\mathbf{p}_{gt,i}$ extracted from the calibration images:

$$L_{rep} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{p}_{pred,i} - \mathbf{p}_{gt,i}\|^2 \quad (12)$$

where N denotes the total number of calibration points across all views. Additionally, to prevent the Spatial Correction Module from producing physically unrealistic 3D deformations, we introduce a spatial regularization term L_{reg} defined as the L_2 norm of the predicted 3D displacement vector $\Delta \mathbf{X}$:

$$L_{reg} = \frac{1}{N} \sum_{i=1}^N \|\Delta \mathbf{X}_i\|^2 \quad (13)$$

The complete training objective is a weighted sum of these two components, formulated as follows:

$$L_{total} = L_{rep} + \lambda L_{reg} \quad (14)$$

In this formulation, the hyperparameter λ controls the trade-off between geometric flexibility and structural rigidity. By minimizing L_{total} using the Adam optimizer, the gradients are backpropagated through the entire differentiable pipeline, allowing TLNet to achieve a global

equilibrium between 3D geometric refinement and 2D residual compensation.

As illustrated in Fig. 4, we compare the proposed TLNet framework with the representative Zhang's and Huang's methods across five fundamental process stages. Traditional analytical paradigms such as Zhang's method rely on ideal geometric assumptions and lack 3D spatial refinement. While Huang's method incorporates target geometry refinement, its ultimate accuracy is restricted by the limited representational capacity of traditional parametric distortion models. In contrast, TLNet achieves logical decoupling by introducing a Differentiable Physics Layer that separates Neural Spatial Refinement from Neural Residual Mapping. This architecture enables a joint end-to-end optimization strategy where 3D geometric fidelity and 2D reprojection accuracy are balanced through global backpropagation within a unified computational graph.

4. Experiments

4.1. Experimental setup

We utilized an industrial camera (IDS UI3370CP-MGL, 1920×1200 pixels [36]) with a set of interchangeable prime lenses for multiple sets of calibration board captures. The focal lengths of the lenses used included 8 mm, 12 mm and 16 mm. Additionally, a cheap camera was employed. The device was a generic USB board camera (model GXIVISION-S2M03, 1920×1080 pixels) equipped with a 1/2.8-inch CMOS sensor ($2.9 \mu\text{m}$ pixel size). The specific hardware configuration and the corresponding checkerboard images captured by these two devices are illustrated in Fig. 5 and Fig. 6 respectively.

The core neural architecture consists of two interconnected multi-layer perceptron modules. The three dimensional spatial correction network contains two hidden layers with 64 neurons each and the two dimensional residual distortion network contains two hidden layers with 256 and 128 neurons respectively. Both internal modules utilize the hyperbolic tangent function Tanh as the activation function. With this configuration, the Spatial Correction Module contains 4611 trainable parameters, while the Distortion Residual Module contains 33,922 trainable parameters. Therefore, the neural network part of TLNet contains 38,533 trainable parameters in total. In addition to these neural parameters, the optimization process also includes four intrinsic parameters, f_x , f_y , c_x , and c_y , as well as seven extrinsic parameters for each calibration view. Thus, for N calibration images, the total number of trainable variables is $38,533 + 4 + 7N$. In the largest experimental setting with 25 images, this number is 38,712, indicating that TLNet remains a lightweight optimization framework.

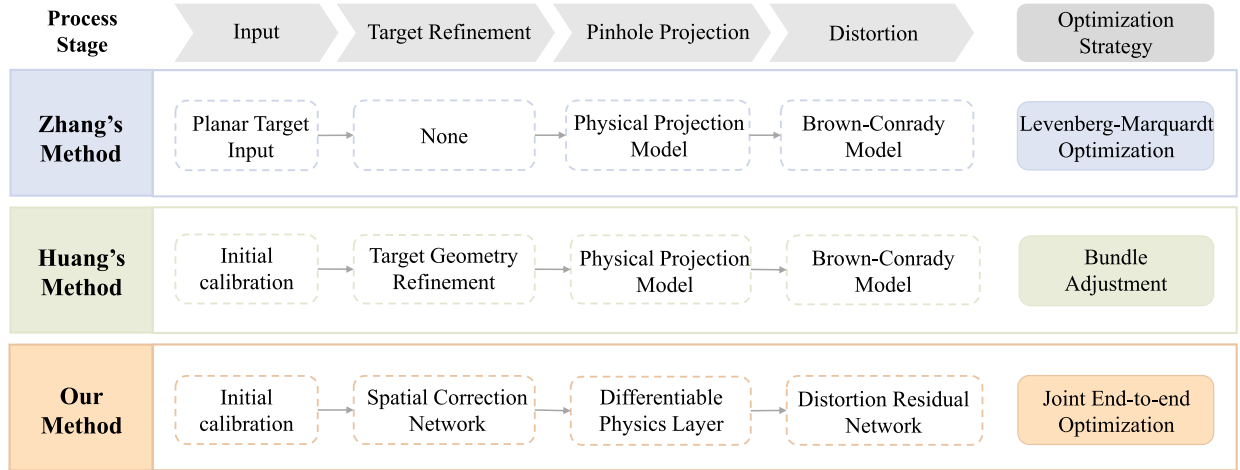
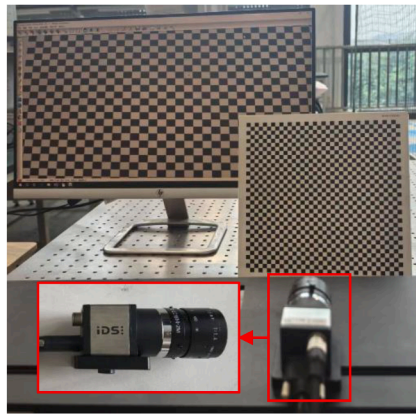
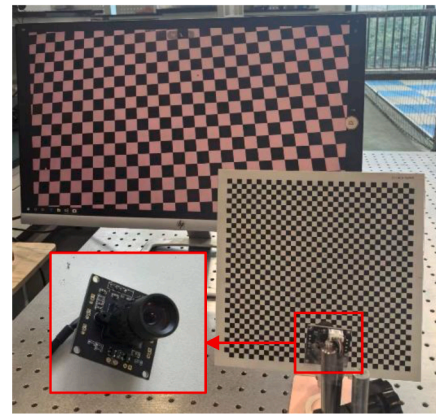


Fig. 4. Comparison of calibration paradigms.

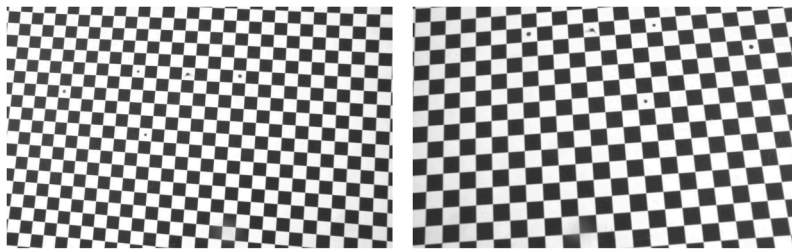


(a) The industrial imaging system

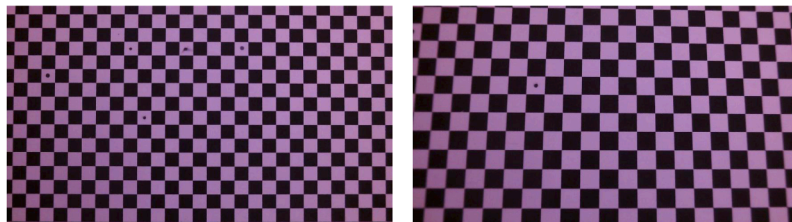


(b) The cheap board camera setup

Fig. 5. Experimental hardware configuration. (a) The industrial imaging system (b) The cheap board camera setup.



(a) Checkerboard images captured by the industrial camera



(b) Checkerboard images captured by the cheap camera

Fig. 6. Original calibration board images. (a) Checkerboard images captured by the industrial camera. (b) Checkerboard images captured by the cheap camera.

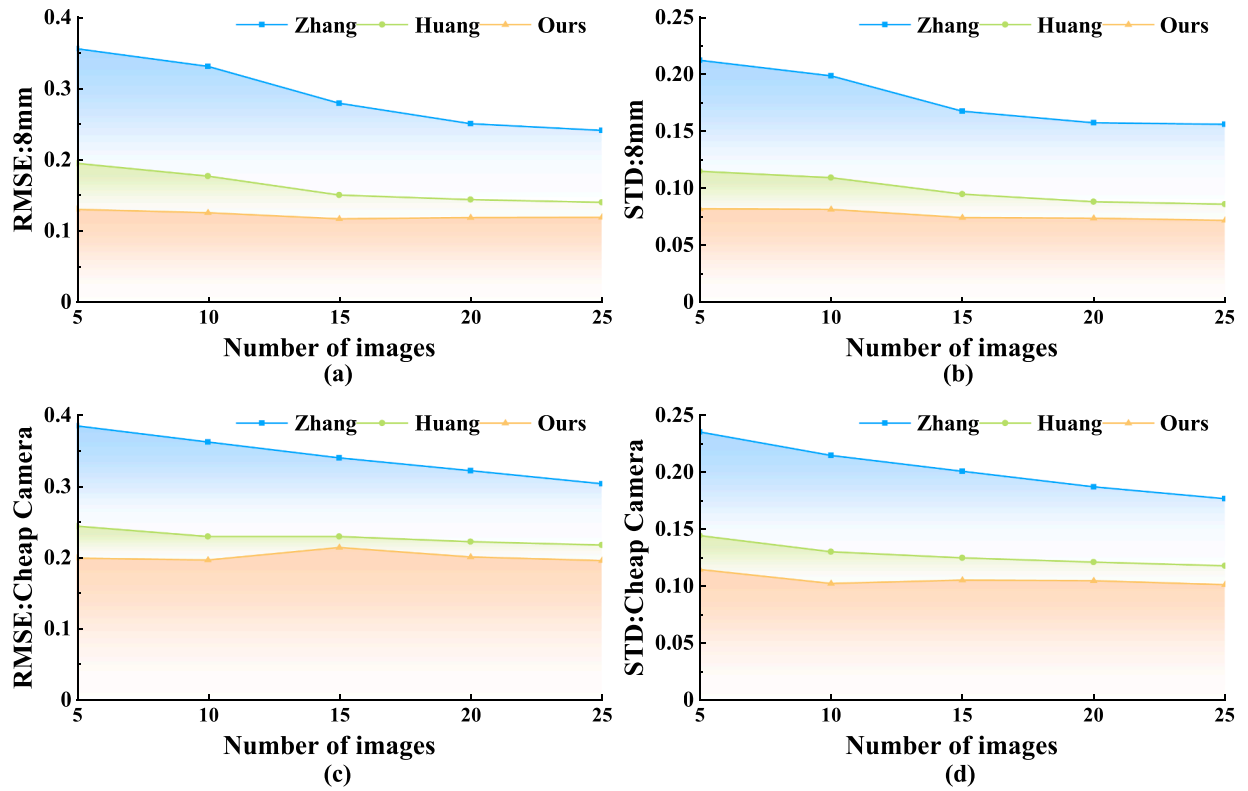


Fig. 7. Comparison of calibration performance under different numbers of images. (a) RMSE for the 8 mm industrial camera. (b) STD for the 8 mm industrial camera. (c) RMSE for the cheap camera. (d) STD for the cheap camera.

Table 1
Comparison of Zhang, Huang and TLNet under different numbers of calibration images.

Camera	Number of Images	RMSE(px)			STD(px)		
		Zhang	Huang	Ours	Zhang	Huang	Ours
Industrial Camera ($f = 8$ mm)	5	0.3555	0.1946	0.1300	0.2120	0.1148	0.0818
	10	0.3309	0.1766	0.1255	0.1985	0.1092	0.0814
	15	0.2788	0.1502	0.1170	0.1675	0.0947	0.0741
	20	0.2503	0.1437	0.1184	0.1573	0.0881	0.0735
	25	0.2409	0.1397	0.1188	0.1559	0.0857	0.0716
Cheap Camera	5	0.3855	0.2446	0.2020	0.2355	0.1446	0.1150
	10	0.3626	0.2301	0.1973	0.2149	0.1304	0.1026
	15	0.3404	0.2301	0.2148	0.2012	0.1252	0.1056
	20	0.3225	0.2231	0.2017	0.1874	0.1215	0.1052
	25	0.3042	0.2181	0.1962	0.1771	0.1181	0.1017

During the joint optimization process the network was trained using the Adam optimizer with a full batch gradient descent strategy. The initial learning rate was set to 0.003 and gradually decayed to $1e-6$ over 80,000 epochs. In our combined loss function the hyperparameter λ which controls the weight of the physical spatial regularization term against the reprojection error was set to 1.0 to ensure the optimal balance between spatial target refinement and geometric distortion compensation.

4.2. Comparison with Zhang and Huang's methods

We compare the proposed TLNet with Zhang's method and Huang's method under different numbers of calibration images and camera settings, including an industrial camera (8mm) and a cheap camera. The quantitative results are summarized in Table 1, and the corresponding trends are illustrated in Fig. 7.

Overall, TLNet achieves the lowest reprojection error across all settings. As shown in Table 1, TLNet outperforms both Zhang and Huang in terms of RMSE and standard deviation, demonstrating improved calibration accuracy and stability.

From the trend curves in Fig. 7, all methods benefit from increasing numbers of calibration images, as expected. However, TLNet shows a clear advantage in the low-data regime. When only a small number of images are available, Zhang's method exhibits higher errors because it assumes a perfectly planar calibration target and does not account for surface deviations in real-world setups, while Huang's method alleviates this issue through joint optimization. In contrast, TLNet maintains relatively low error even with limited data, indicating better generalization and robustness.

Moreover, TLNet is less sensitive to the number of calibration images compared to the other methods. While the performance of Zhang and Huang improves as more images are used, TLNet exhibits stable performance across different data regimes. This suggests that the proposed method can achieve reliable calibration even with a limited number of observations.

In addition, TLNet produces lower standard deviation compared to the other methods, suggesting more stable calibration results across different image sets. This stability can be attributed to the unified modeling of geometric correction and residual distortion, which reduces sensitivity to local noise and modeling errors.

These results demonstrate that TLNet not only improves calibration accuracy but also enhances robustness under varying data conditions, especially when the planar assumption is violated or the number of calibration images is limited.

4.3. Robustness to target deformation

In our physical experiments, we utilized a glass substrate calibration board to establish a near-ideal planar reference. However, practical applications often rely on less rigid and lower-cost alternatives.

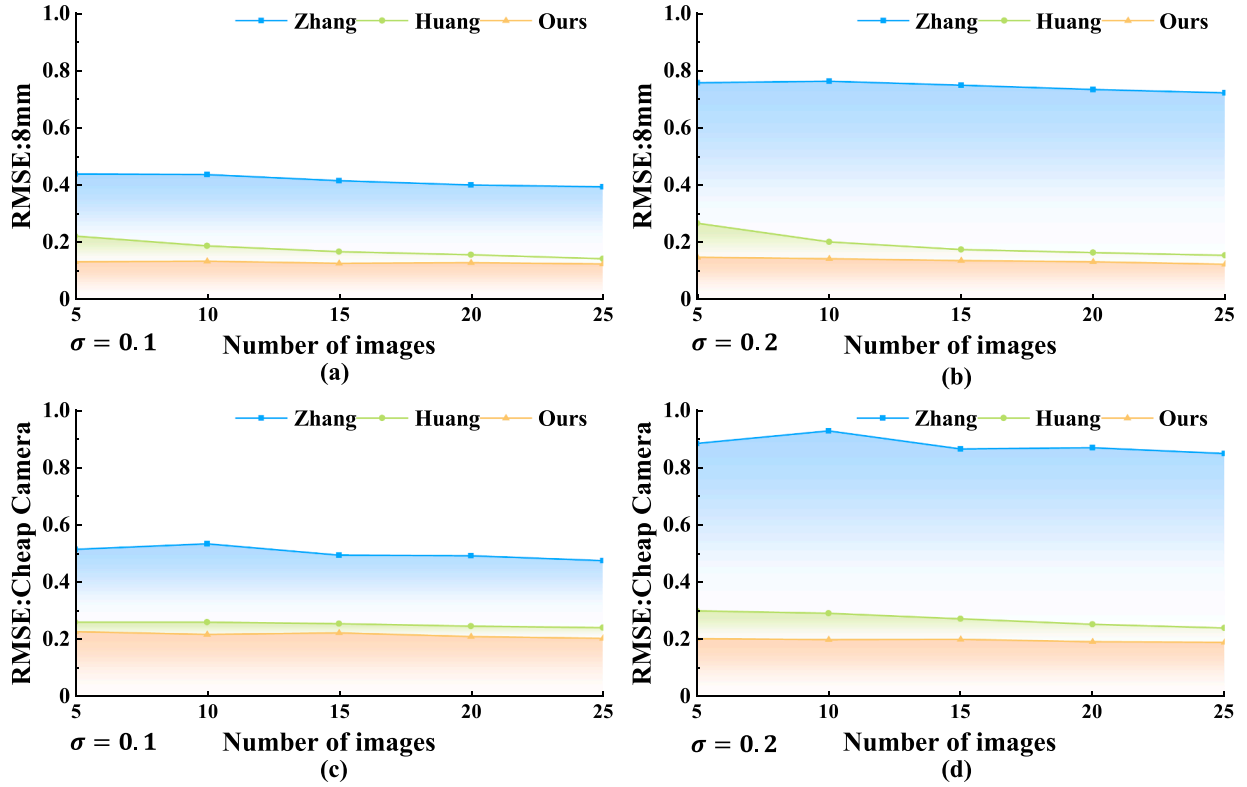


Fig. 8. Comparison of reprojection RMSE under different levels of noise. (a) Performance on the 8 mm industrial camera with $\sigma = 0.1$ mm. (b) Performance on the 8 mm industrial camera with $\sigma = 0.2$ mm. (c) Results for the cheap camera with $\sigma = 0.1$ mm. (d) Results for the cheap camera with $\sigma = 0.2$ mm.

To evaluate the robustness of TLNet against such non-ideal conditions, we introduced Gaussian noise to the Z -coordinates of the checkerboard corners. The perturbation was applied in the world coordinate system to simulate out-of-plane deformation of the calibration board. Specifically, for each nominal checkerboard corner $\mathbf{P}_j = [X_j, Y_j, Z_j]^T$, the perturbed point was defined as

$$\mathbf{P}'_j = \begin{bmatrix} X_j \\ Y_j \\ Z_j + \epsilon_j \end{bmatrix}, \quad \epsilon_j \sim \mathcal{N}(0, \sigma^2). \quad (15)$$

The perturbation was applied only along the Z_w -direction, while the X_w - and Y_w -coordinates remained unchanged. In each trial, the same perturbed target geometry was shared by all calibration views, corresponding to a fixed physically deformed calibration board. This mathematical degradation simulates the severe out-of-plane warping and manufacturing inaccuracies found in non-glass targets such as printed paper or computer monitors. We set the noise levels at $\sigma = 0.1$ mm and 0.2 mm along the Z axis to observe how each method reacts to increasing structural deformation. Table 2 lists the precise numerical results, while Fig. 8 illustrates the overall error trends across different camera setups.

The experimental results demonstrate that TLNet performs best under all noise conditions because it maintains the lowest RMSE and shows stability. As shown in Fig. 8, the performance of Zhang's method drops significantly as the board becomes less flat. For the 8 mm camera at $\sigma = 0.2$ mm, the reprojection error reaches 0.75 pixels because the model assumes the board is a perfect plane. Since this classic assumption cannot explain 3D geometric errors, the entire calibration model fails to provide accurate results. Huang's method helps by optimizing the target shape but its accuracy still fluctuates a lot when the noise level increases.

TLNet is much more accurate and maintains the highest level of stability among the three methods. The error curves for TLNet stay flat regardless of the noise level or the number of calibration images used

Table 2

Comparison of reprojection RMSE (px) under different noise levels (σ).

Camera	σ (mm)	Method	Number of Images				
			5	10	15	20	25
Industrial Camera (8 mm)	0.1	Zhang	0.4370	0.4357	0.4135	0.3987	0.3920
		Huang	0.2207	0.1860	0.1663	0.1556	0.1407
		Ours	0.1301	0.1327	0.1256	0.1268	0.1233
	0.2	Zhang	0.7560	0.7612	0.7477	0.7325	0.7204
		Huang	0.2652	0.1997	0.1733	0.1625	0.1532
		Ours	0.1469	0.1414	0.1343	0.1301	0.1223
Cheap Camera	0.1	Zhang	0.5141	0.5333	0.4940	0.4921	0.4750
		Huang	0.2602	0.2596	0.2546	0.2455	0.2410
		Ours	0.2263	0.2176	0.2226	0.2101	0.2032
	0.2	Zhang	0.8850	0.9288	0.8653	0.8696	0.8493
		Huang	0.3195	0.3112	0.2913	0.2724	0.2597
		Ours	0.2225	0.2184	0.2195	0.2116	0.2094

in the process. This stability proves that our Spatial Correction Module models 3D warping and effectively separates geometric errors from the projection process. These findings confirm that TLNet provides reliable calibration results even when the physical board violates the ideal planar assumption.

The performance between TLNet and Huang's method highlights a critical limitation of traditional distortion models. Although Huang's method considers board errors, it still relies on old physical formulas that lack the flexibility to fix complex errors. TLNet uses a two-step correction strategy to handle different types of deviations. First, the Spatial Correction Module handles the large 3D bends of the calibration board. Then the Distortion Residual Module acts as a strong tool to fix any remaining projection errors on the image plane. Even with high noise, TLNet achieves lower errors than Huang because neural networks can catch complex system errors that pure physical models cannot reach.

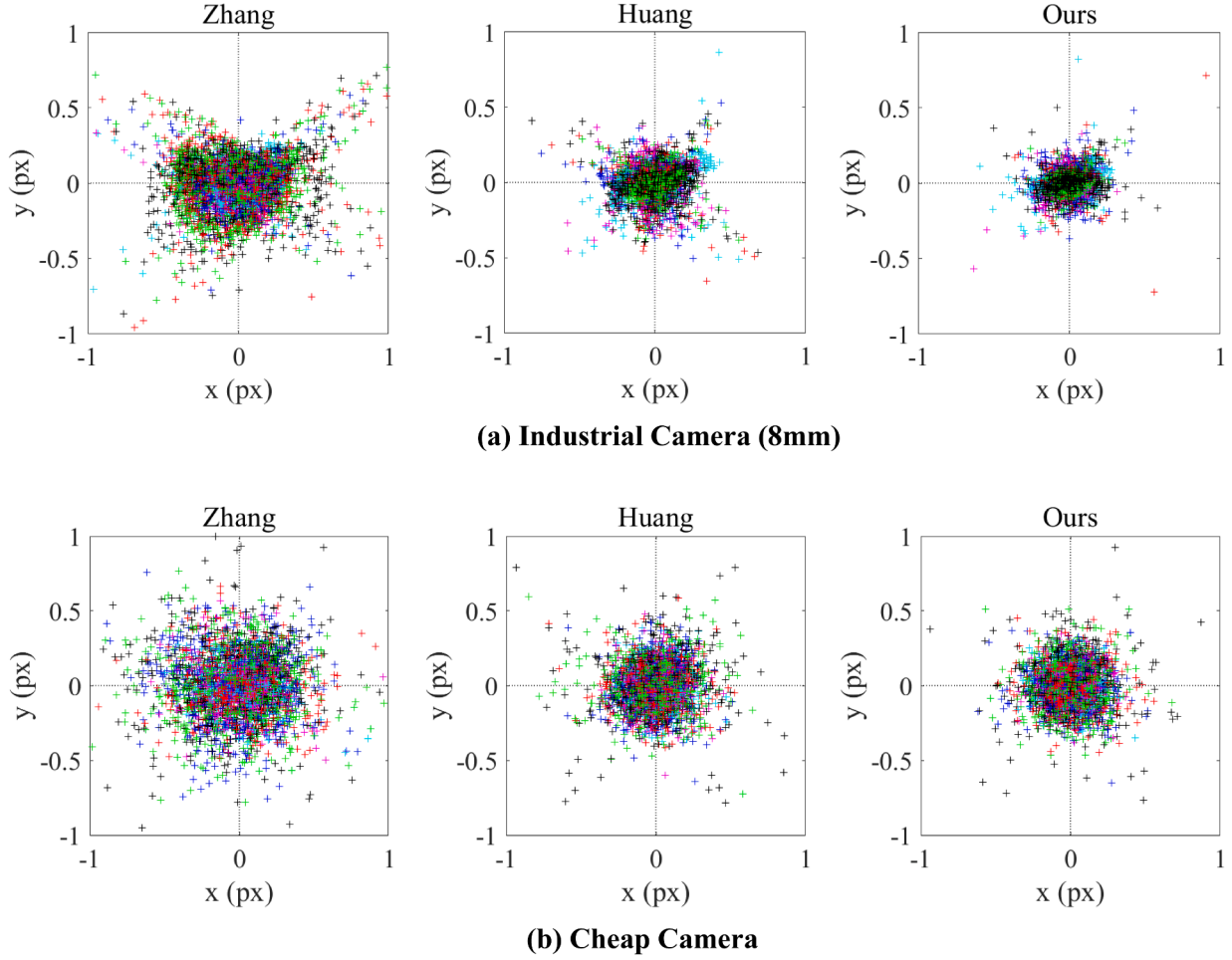


Fig. 9. Reprojection error scatter plots (px) comparing Zhang's method, Huang's method, and our proposed method across two camera systems: (a) Industrial Camera with an 8 mm lens, and (b) a Cheap Camera with significant non-linear distortion.

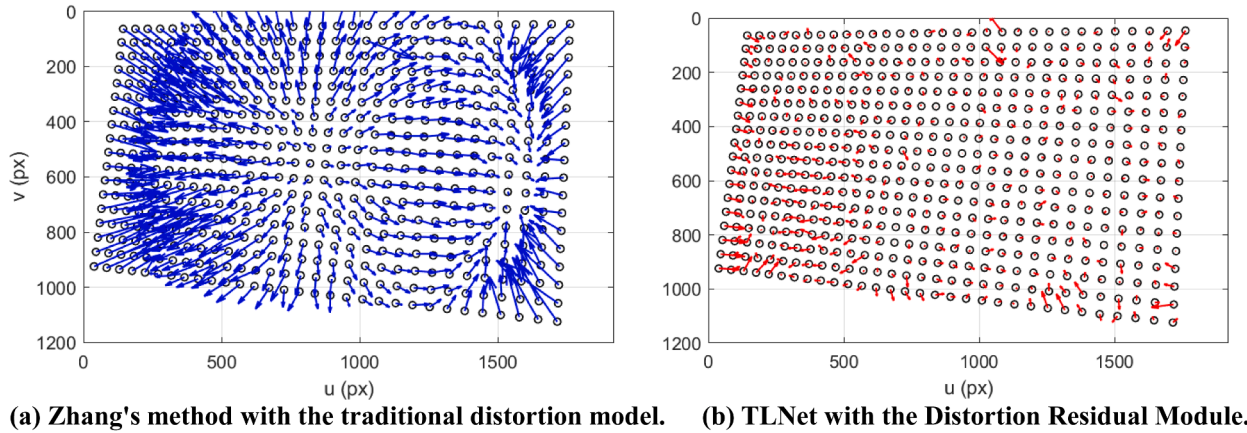


Fig. 10. Visualization of actual reprojection residual vectors on a representative calibration image. (a) Zhang's method with the traditional radial/tangential distortion model. (b) TLNet with the Distortion Residual Module. The arrows are magnified by a factor of 300 for visualization.

4.4. Error distribution and spatial analysis

In this section, all experimental results are derived from a test set of 10 independent images for each camera system to ensure statistical consistency. The performance is assessed across an industrial camera (8 mm lens) and a Cheap Camera.

The scatter plots in Fig. 9 show the distribution of reprojection residuals (dx, dy) for all test points. For the industrial camera, both optimized methods perform better than the Zhang baseline. However, Huang's results still show more dispersion than our proposed method. TLNet achieves the most compact residual cluster because the vast majority of points stay tightly centered at the origin. Even in the Cheap Camera

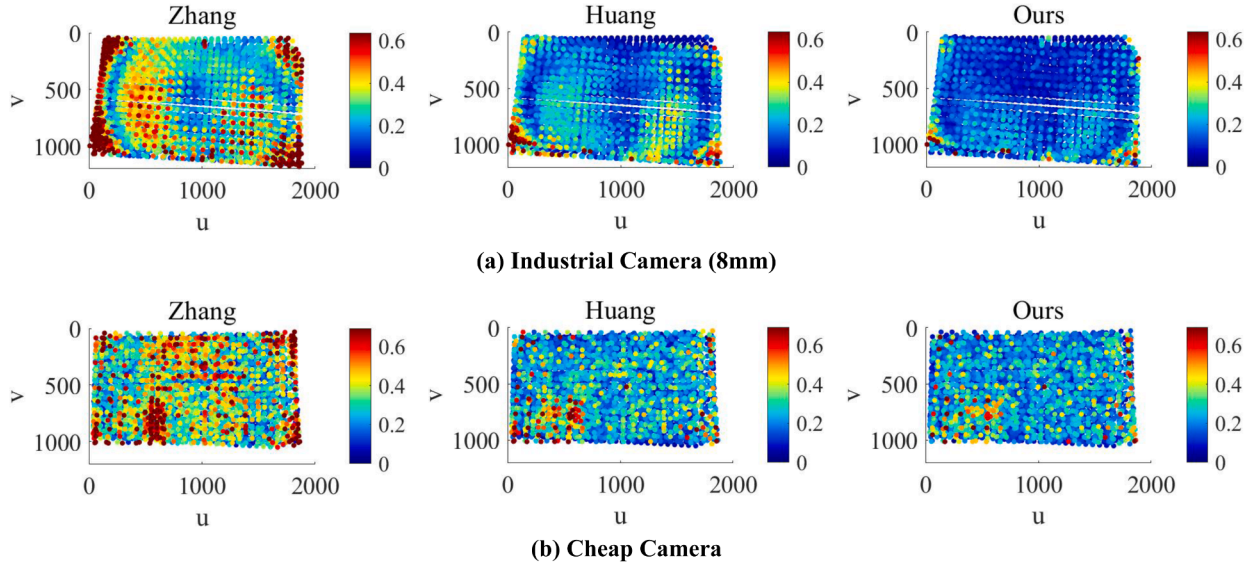


Fig. 11. Spatial error heatmaps across the $u-v$ image plane for (a) the Industrial Camera (8mm lens) and (b) the Cheap Camera, based on 10 independent test images.

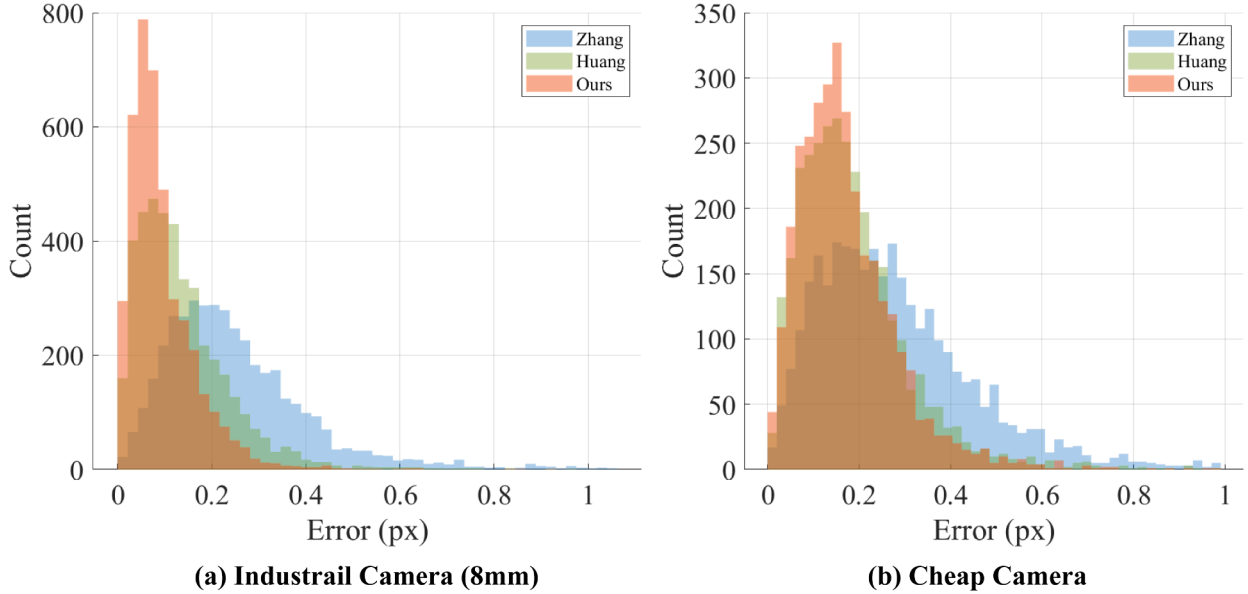


Fig. 12. Reprojection error distribution histograms for (a) the industrial camera and (b) the Cheap Camera based on 10 independent test images. The proposed method exhibits a narrower and more centered profile compared to the baseline methods.

setup, our method maintains high centering precision. It keeps the residuals within a small radius around $(0,0)$ despite the lower-quality optics.

To further improve the interpretability of the MLP-based residual compensation, we visualize the actual reprojection residual vectors on a representative calibration image, as shown in Fig. 10. The visualization is performed using the industrial camera with the 8 mm lens under the 10-image calibration setting. The detected checkerboard corners are used as reference points, and the arrows indicate the residual vectors from the observed image points to the predicted image points. Compared with Zhang's method, which uses the traditional radial/tangential distortion model, TLNet produces shorter and more spatially uniform residual vectors. This provides a more intuitive visualization of the effect of the MLP-based residual compensation. It also suggests that the Distortion Residual Module improves the representation of residual image-plane errors that are difficult to fully describe using a limited

number of fixed distortion coefficients. The visualization therefore improves interpretability by showing where and how the residual errors are reduced in the image plane.

The spatial heatmaps in Fig. 11 illustrate the error magnitude across the $u-v$ plane. For the industrial camera, Zhang results exhibit prominent high error hotspots at the corners. Huang method reduces these localized peaks but still maintains several visible high error clusters. In contrast, our method produces a near uniform blue landscape and eliminates the systematic hotspots observed in the baseline results. For the Cheap Camera setup, our method provides a low and stable error profile across the entire sensor surface. This approach effectively smooths out localized error peaks and maintains superior calibration precision despite the lower quality optics.

Fig. 12 uses error histograms to quantify the statistical reliability of the results. For the industrial camera, the error frequency of our method peaks sharply near zero pixels. The distribution for TLNet is much

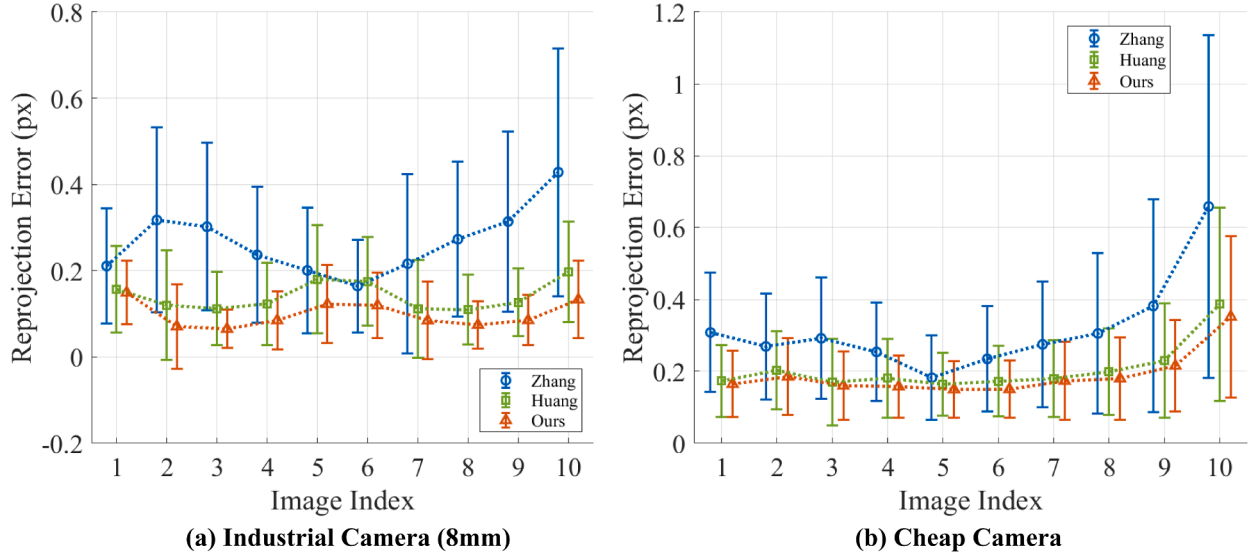


Fig. 13. Per-view reprojection error statistics for (a) the industrial camera and (b) the cheap camera.

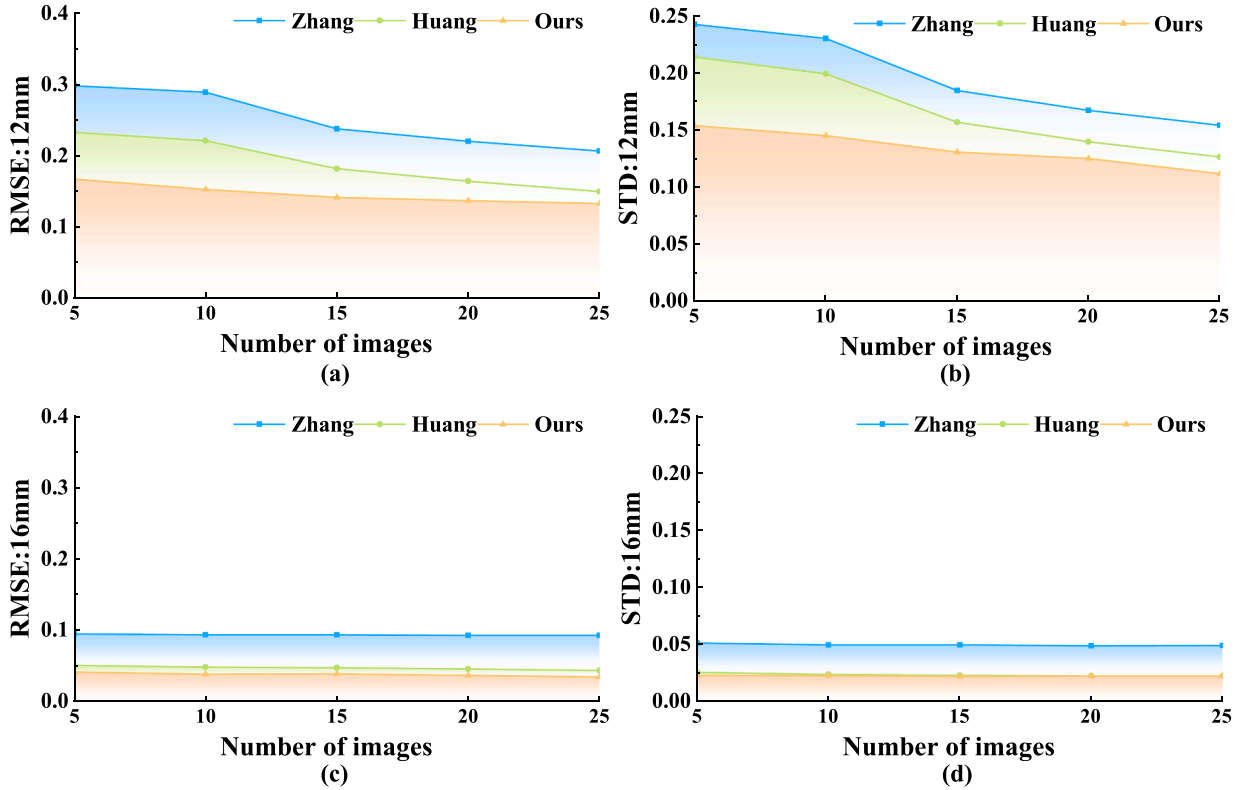


Fig. 14. Comparison of calibration performance under different numbers of images for industrial cameras equipped with 12mm and 16mm lenses: (a) RMSE and (b) STD trends for the 12mm lens; (c) RMSE and (d) STD trends for the 16mm lens.

narrower than those of Zhang and Huang. This indicates a higher concentration of low-magnitude errors. In the Cheap Camera setup, the error range expands for all methods due to optical limits. However, TLNet still maintains a superior distribution with a higher peak frequency in the low-error range and a faster decay as errors increase.

Finally, Fig. 13 demonstrates the stability of TLNet across different camera poses. For the industrial camera, our method maintains the lowest mean reprojection error across all 10 images. The standard deviations for TLNet are smaller than those of Zhang and Huang, which indicates high reliability for each view. In the Cheap Camera setup, our method provides the most stable performance. The narrow error bars

confirm that our calibration accuracy remains robust even when using low-cost hardware.

4.5. Stability test

To further evaluate the algorithmic stability and consistency of the proposed framework, we extend our experiments to include industrial cameras equipped with 12mm and 16mm focal length lenses. Testing across these diverse optical profiles is crucial to confirm that TLNet provides a stable calibration solution, demonstrating that it does not merely overfit to specific extreme distortion patterns.

Table 3

Comparison of Zhang, Huang and TLNet under different numbers of calibration images for industrial cameras equipped with 12 mm and 16 mm lenses.

Camera	Number of Images	RMSE(px)			STD(px)		
		Zhang	Huang	Ours	Zhang	Huang	Ours
Industrial Camera ($f = 12$ mm)	5	0.2979	0.2322	0.1669	0.2424	0.2138	0.1538
	10	0.2888	0.2207	0.1521	0.2303	0.1992	0.1450
	15	0.2373	0.1812	0.1410	0.1845	0.1568	0.1305
	20	0.2200	0.1637	0.1364	0.1671	0.1397	0.1248
	25	0.2061	0.1492	0.1328	0.1540	0.1261	0.1115
Industrial Camera ($f = 16$ mm)	5	0.0951	0.0508	0.0416	0.0513	0.0258	0.0231
	10	0.0936	0.0483	0.0386	0.0498	0.0237	0.0226
	15	0.0938	0.0473	0.0389	0.0496	0.0231	0.0223
	20	0.0927	0.0459	0.0370	0.0489	0.0225	0.0224
	25	0.0927	0.0436	0.0345	0.0492	0.0225	0.0225

The quantitative data in Table 3 reveals the absolute accuracy advantage of the proposed framework.

In both configurations, TLNet achieves the lowest root mean square error and standard deviation regardless of the dataset size. When the calibration dataset is limited, such as using only 5 images, traditional methods experience a sharp degradation in the 12 mm setup. For example, Zhang's error surges to nearly 0.3 pixels. TLNet suppresses this error spike, maintaining a stable error of approximately 0.16 pixels. For the 16 mm lens, where inherent optical distortions are milder, the baseline errors for all methods decrease. Even in this scenario, TLNet secures the lowest absolute error without introducing any negative structural bias, demonstrating its superior precision and data efficiency.

The performance trends illustrated in Fig. 14 further demonstrate the algorithmic stability of TLNet compared to discrete optimization strategies. In the 12 mm configuration shown in Fig. 14(a) and (b), the error curves for Zhang's and Huang's methods exhibit noticeable fluctuations and a strong dependency on the volume of calibration images. Conversely, the curve for TLNet remains flat and robust even in low-data regimes, proving its ability to converge without dense multi-view redundancy. In the 16 mm configuration shown in Fig. 14(c) and (d), the performance curves of TLNet maintain a high degree of stability with negligible fluctuation. This behavior confirms that the neural components scale their compensation intensity based on the physical reality, effectively coordinating the spatial and residual networks to prevent convergence to local optima.

5. Conclusion

In this paper we proposed a unified neural and physical framework named TLNet to address the fundamental accuracy bottlenecks in classic camera calibration. Traditional calibration paradigms struggle with geometric deviations caused by imperfect planar targets and the modeling limitations of parametric distortion. To overcome these physical constraints TLNet introduces an end to end joint optimization mechanism. This framework simultaneously models and compensates for three dimensional target deformation and two dimensional residual distortion.

Comprehensive experimental evaluations validate the superiority and robustness of our framework. The results demonstrate that TLNet maintains high calibration accuracy even when the number of available images is limited. When using low-precision calibration targets, the network exhibits robust stability against severe out-of-plane structural warping. Moreover, spatial analyses reveal that TLNet eliminates localized error hotspots to produce a uniform error distribution across the image plane. Finally, tests across diverse optical setups confirm that our method adapts to different focal lengths and lens qualities without over-fitting, ensuring stable calibration performance. Overall, TLNet offers a robust and highly accurate geometric calibration solution for practical 3D vision applications.

Disclosures

The authors declare no conflicts of interest.

CRediT authorship contribution statement

Kelei Wang: Writing – original draft, Validation, Data curation; **Junpeng Xue:** Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization; **Feifei He:** Validation, Software; **Changzhi Yu:** Validation, Methodology, Investigation, Data curation; **Jiashi Zeng:** Visualization, Investigation; **Yusong Meng:** Visualization, Investigation; **Ran Jia:** Methodology, Formal analysis.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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